PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

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Program/Batch	B084-Networkwalks
Date	13 November 2026
Modules completed	W2-PM4 (Footprinting with theHarvester) W2-PM5 (Zenmap Scanning)
Client/Target	1. Microsoft – Assigned theHarvester reconnaissance exercise. 2. Own local LAN – Network discovery using Zenmap.
Permission secured from client?	Yes
Phases covered	**Phase 1:** Reconnaissance and Footprinting – Completed. **Phase 2:** Scanning and Network Discovery – Completed. **Phases 3–5:** Not performed as part of the completed Week 2 practical activities.

1. Liability Disclaimer

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2. Introduction

This report documents the practical activities completed during Week 2 of the Networkwalks Cybersecurity Internship. The work focused on footprinting, reconnaissance and network discovery using theHarvester and Zenmap.

The exercises involved collecting publicly available information through reconnaissance tools and identifying active devices on a local network through network scanning.

The objective was to understand how information gathering and network discovery contribute to the initial phases of a penetration test, and how the results can be documented for security assessment.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux	Environment for reconnaissance exercises
Ubuntu Linux	Operating system used for network scanning
theHarvester 4.11.1	Public-source information gathering
Zenmap	Graphical interface for Nmap
Nmap 7.94+SVN	Network discovery and host identification
Graphical topology export	Network topology documentation

4. Activities Performed

4.1 W2-PM4 – Footprinting with theHarvester

Objective

To understand how theHarvester collects publicly available information associated with a domain through different search engines and data sources.

Task 1 – Baidu Search

The first exercise involved using Baidu as the information source.

Command used:

```
theHarvester -d microsoft.com -l 1000 -b baidu -f task1_baidu
```

The exercise was used to understand how the tool searches publicly indexed information associated with a specified domain.

<theHarvester>

<cmd>-d microsoft.com -l 1000 -b baidu -f task1_baidu</cmd>

</theHarvester>

Results:

- **Target domain:** microsoft.com
- **Source:** Baidu
- **Email addresses found:** 1
- **IP addresses found:** 0
- **Hosts found:** 11
- **People found:** 0

Email address identified:

- `user@contoso.onmicrosoft.com`

Hostnames discovered:

1. 2fsupport.microsoft.com
2. account.microsoft.com
3. accountprotection.microsoft.com
4. c.urs.microsoft.com
5. copilot.microsoft.com
6. developer.microsoft.com
7. hxd.research.microsoft.com
8. learn.microsoft.com
9. prod.support.services.microsoft.com
10. support.microsoft.com
11. ti.defender.microsoft.com

Observation: The Baidu source returned 11 hostnames and one email address. No IP addresses or people were identified in this search.

Task 2 – All Available Sources

The second exercise involved running theHarvester against all configured sources supported by the installed version.

Command used:

```
theHarvester -d microsoft.com -l 50 -b all -f task2_all
```

The purpose was to understand how multiple information sources can contribute to domain footprinting.

Results:

Finding	Count
Email addresses identified	3
Host entries collected	10,083
Host entries containing IP information	640
Distinct IP addresses identified	606
Distinct hostnames	9,565
People identified	0

Observation:

TheHarvester collected a large set of publicly available domain-related information from multiple sources. The results included email addresses, hostnames and IP address associations. This exercise demonstrated how aggregated OSINT sources can provide a broad footprint of a target domain.

4.2 Network Scanning with Zenmap

Objective

To identify active devices within my local network, record their IP and available MAC addresses, and generate a network topology.

Environment

- Operating system: Ubuntu Linux
- Application: Zenmap
- Scanner: Nmap
- Scan profile: Ping scan

Step 1 – Network Identification

The local network configuration was identified using Ubuntu network configuration utilities.

The subnet used for the scan was:

`10.137.209.0/24`

Step 2 – Host Discovery

Zenmap was configured with the identified subnet as the target and the Ping Scan profile was selected.

The corresponding Nmap command was:

```
nmap -sn 10.137.209.0/24
```

The scan checked 256 IP addresses and identified three live hosts.

Step 3 – Scan Results

Number of live hosts discovered: **3**

No.	IP Address	MAC Address	Observation
1	10.137.209.41	92:A7:90:A9:E7:72	Default gateway
2	10.137.209.76	0A:27:32:C5:EA:3A	Discovered host
3	10.137.209.62	Not displayed	My Ubuntu laptop

The MAC address of the third device was not returned in the scan output.

Step 4 – Network Topology

After completing host discovery, I opened the Topology tab in Zenmap and reviewed the discovered network devices.

The topology was saved in PDF format for inclusion in the final report.

Output file: network_topology.pdf

The scan results and topology provide a visual and technical record of the active devices discovered within the scanned subnet.

5. Risk Analysis and Impact

The following observations were considered based on the activities performed.

#	Evidence / Observation	Potential Security Implication	Assessment
1	Publicly indexed domain information	May provide information useful for targeted reconnaissance	Requires review
2	Publicly visible email addresses, if returned	May increase exposure to phishing attempts	Requires review
3	Multiple live devices on the local network	Unexpected devices may require identification	Requires verification
4	Gateway and device MAC information	Helps identify devices and network infrastructure	Informational

These observations are not confirmed vulnerabilities. The exercises focused on information gathering and host discovery. No exploitation or vulnerability validation was performed.

6. Recommendations

Review publicly exposed information: Organizations should periodically assess what domain-related information is publicly indexed.

1. **Protect organizational email addresses:** Appropriate email security controls and phishing awareness practices should be maintained.
2. **Maintain an accurate device inventory:** Organizations should identify and document devices connected to their networks.
3. **Investigate unexpected devices:** Unknown hosts discovered during network scanning should be verified by the network administrator.
4. **Monitor network infrastructure:** Gateways and other network devices should be properly configured and monitored.
5. **Maintain network documentation:** Network topology and device records should be updated regularly.
6. **Perform authorized security assessments:** Reconnaissance and network scanning should always remain within the approved testing scope.

7. Conclusion

During Week 2 of the Networkwalks Cybersecurity Internship, I completed practical exercises covering footprinting, reconnaissance and network scanning.

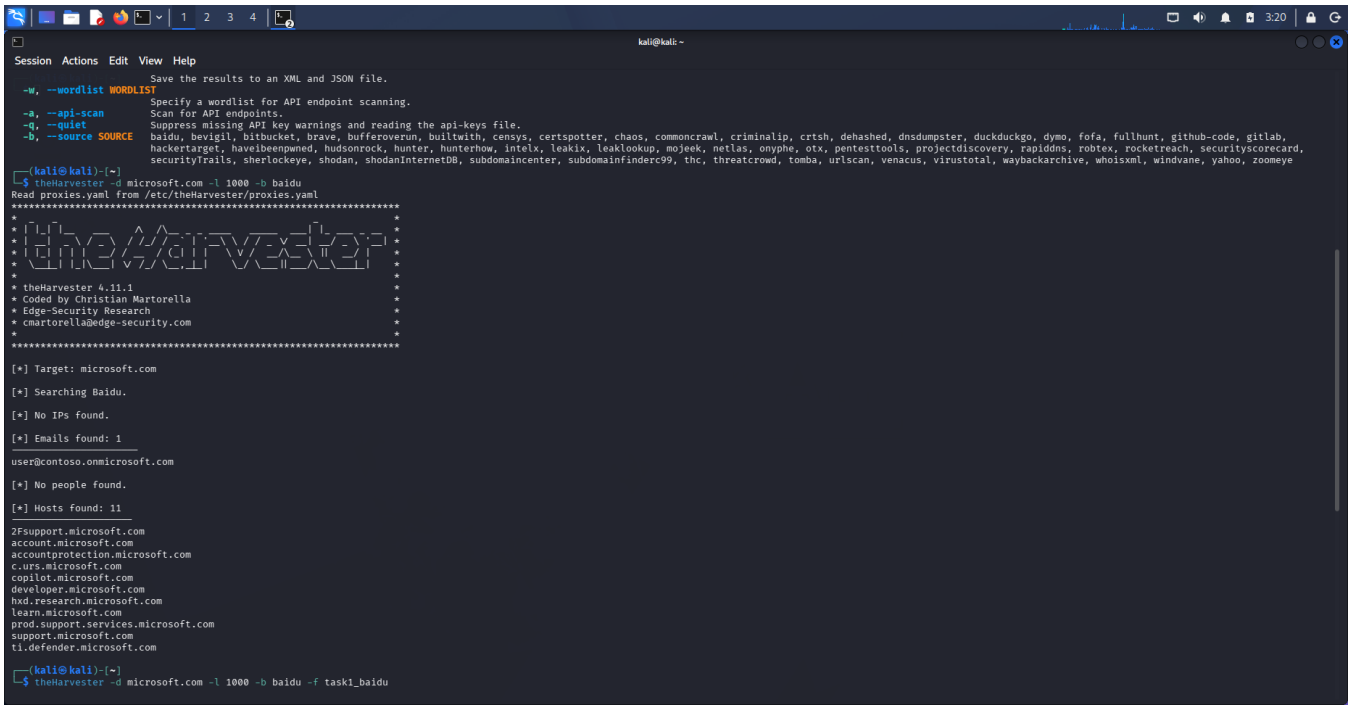
Using theHarvester, I explored how publicly available information can be collected from different sources during the initial reconnaissance phase. The exercises helped me understand the role of search engines and external information sources in building a domain footprint.

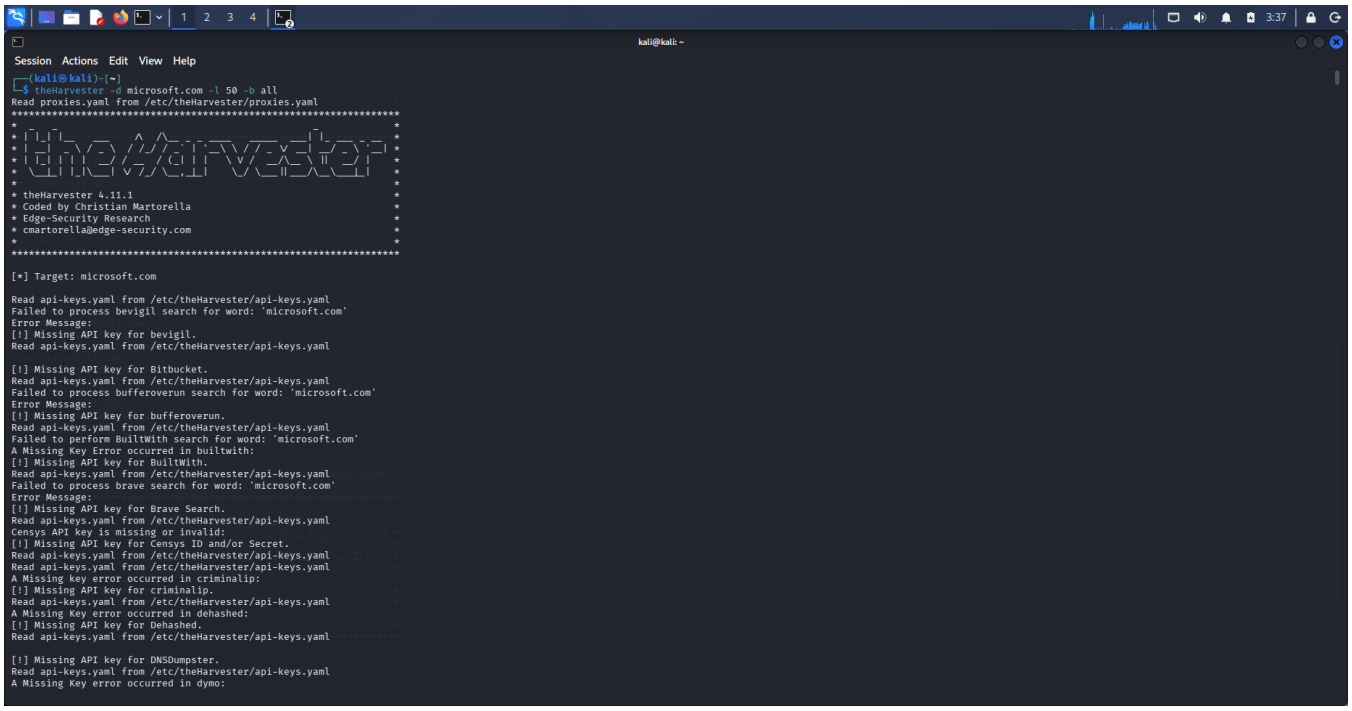
Using Zenmap, I performed a Ping Scan against my local network. The scan covered 256 IP addresses and identified three live hosts. I examined the available IP and MAC address information and generated a network topology in PDF format.

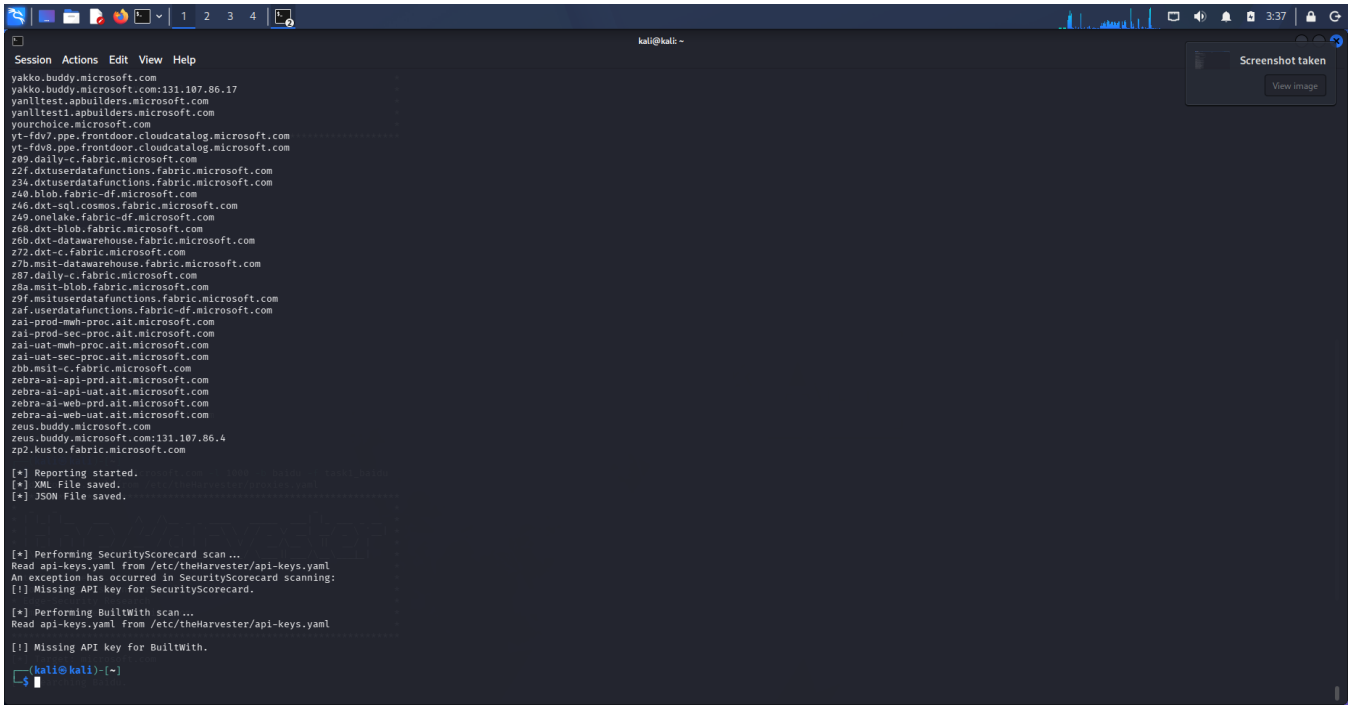
These exercises helped me understand the importance of reconnaissance and network discovery in cybersecurity assessments. They also highlighted the importance of interpreting findings carefully, documenting actual results and distinguishing information exposure from confirmed vulnerabilities.

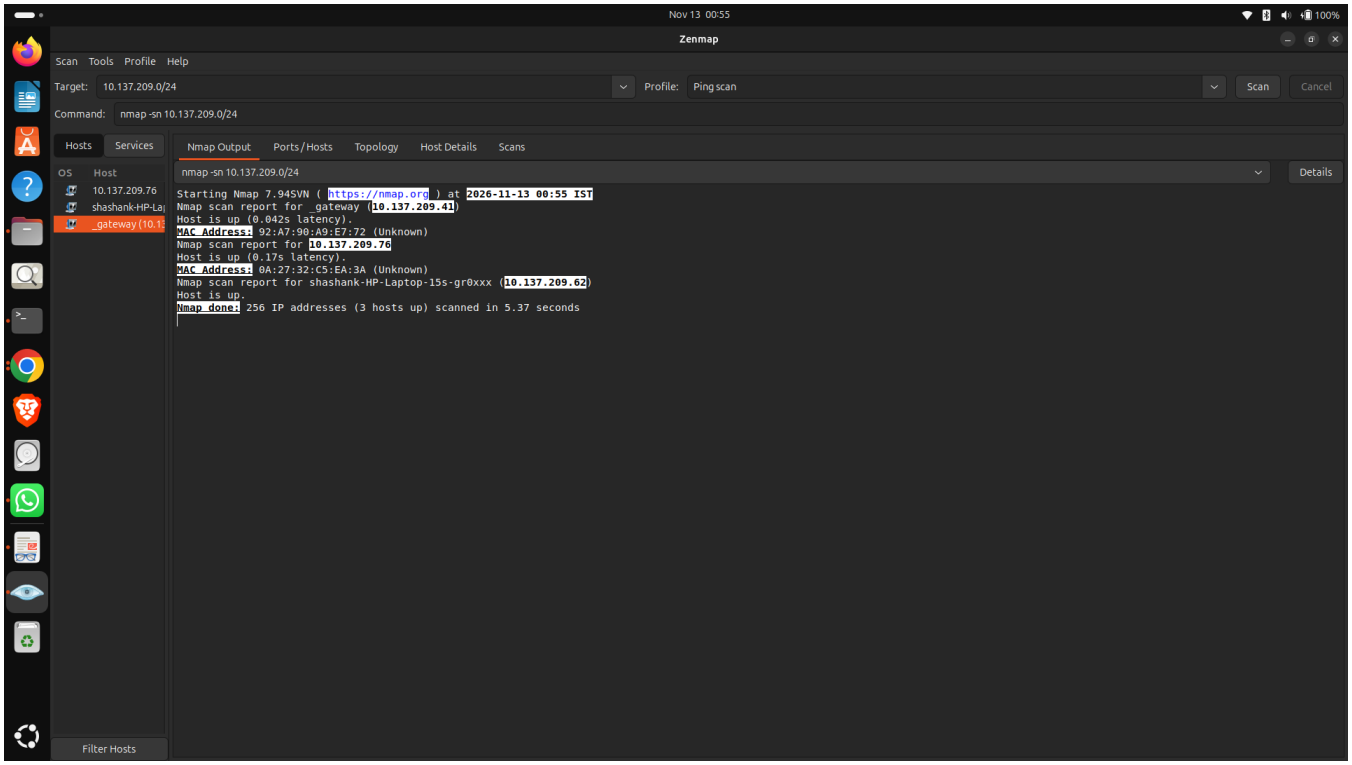
8. Evidences Collected

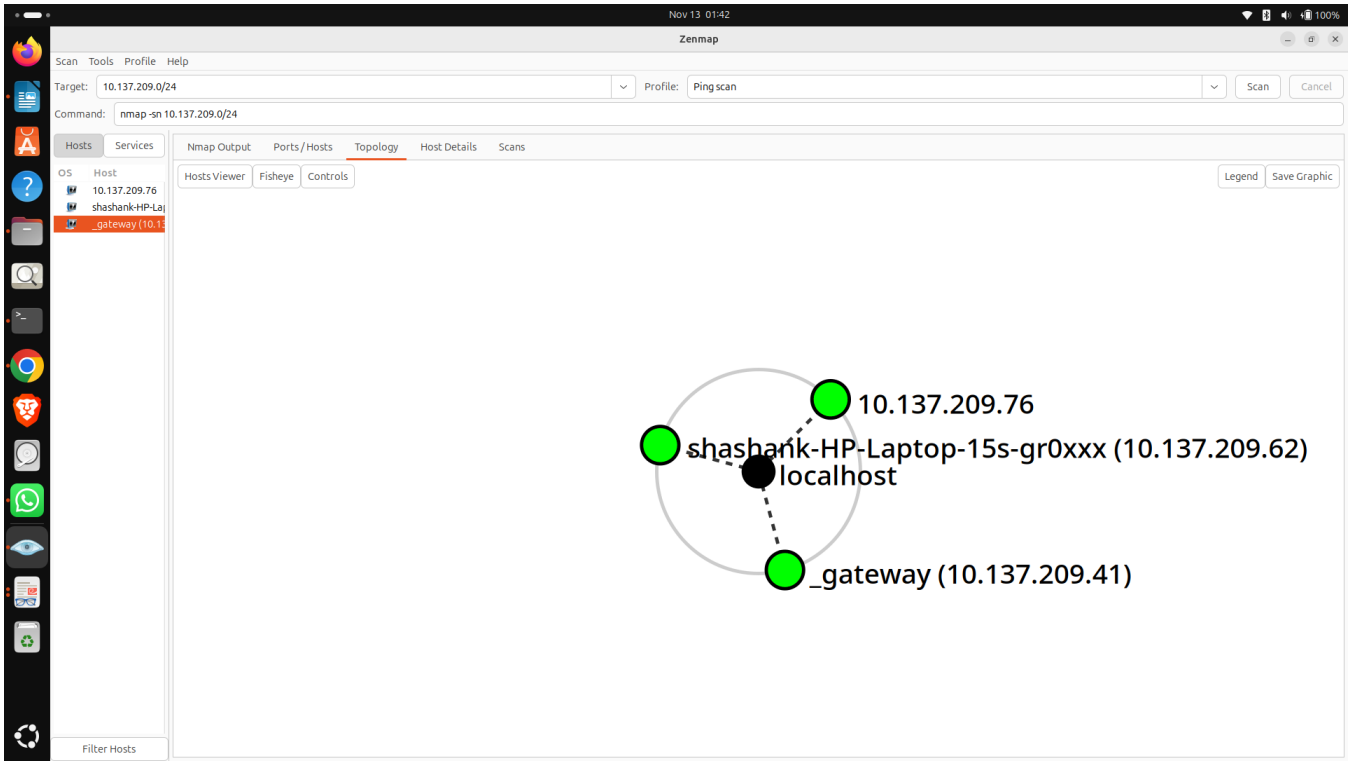





kali@kali: ~
Session Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ theHarvester -d microsoft.com -l 50 -b all
Read proxies.yaml from /etc/theHarvester/proxies.yaml
* theHarvester 4.11.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
[*] Target: microsoft.com
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bevigil search for word: 'microsoft.com'
Error Message:
[!] Missing API key for bevigil.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
[!] Missing API key for Bitbucket.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bufferoverun search for word: 'microsoft.com'
Error Message:
[!] Missing API key for bufferoverun.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to perform BuiltWith search for word: 'microsoft.com'
A Missing Key Error occurred in builtwith:
[!] Missing API key for BuiltWith.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process brave search for word: 'microsoft.com'
Error Message:
[!] Missing API key for Brave Search.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Censys API key is missing or invalid:
[!] Missing API key for Censys ID and/or Secret.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing key error occurred in criminalip:
[!] Missing API key for criminalip.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing Key error occurred in dehashed:
[!] Missing API key for Dehashed.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
[!] Missing API key for DNSDumpster.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing Key error occurred in dymo:


kali@kali: ~
Session Actions Edit View Help
Screenshot taken
View image
yakko.buddy.microsoft.com
yakko.buddy.microsoft.com:131.107.86.17
yanlltest.apbuilders.microsoft.com
yanlltest1.apbuilders.microsoft.com
yourchoice.microsoft.com
yt-fdv7.ppe.frontdoor.cloudcatalog.microsoft.com
yt-fdv8.ppe.frontdoor.cloudcatalog.microsoft.com
z09.daily-c.fabric.microsoft.com
z2f.dxtuserdatafunctions.fabric.microsoft.com
z34.dxtuserdatafunctions.fabric.microsoft.com
z40.blob.fabric-df.microsoft.com
z46.dxt-sql.cosmos.fabric.microsoft.com
z49.onelake.fabric-df.microsoft.com
z68.dxt-blob.fabric.microsoft.com
z6b.dxt-datawarehouse.fabric.microsoft.com
z72.dxt-c.fabric.microsoft.com
z7b.msit-datawarehouse.fabric.microsoft.com
z87.daily-c.fabric.microsoft.com
z8a.msit-blob.fabric.microsoft.com
z9f.msituserdatafunctions.fabric.microsoft.com
zaf.userdatafunctions.fabric-df.microsoft.com
zai-prod-mwh-proc.ait.microsoft.com
zai-prod-sec-proc.ait.microsoft.com
zai-uat-mwh-proc.ait.microsoft.com
zai-uat-sec-proc.ait.microsoft.com
zbb.msit-c.fabric.microsoft.com
zebra-ai-api-prd.ait.microsoft.com
zebra-ai-api-uat.ait.microsoft.com
zebra-ai-web-prd.ait.microsoft.com
zebra-ai-web-uat.ait.microsoft.com
zeus.buddy.microsoft.com
zeus.buddy.microsoft.com:131.107.86.4
zp2.kusto.fabric.microsoft.com
[*] Reporting started.
[*] XML File saved.
[*] JSON File saved.
[*] Performing SecurityScorecard scan...
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
An exception has occurred in SecurityScorecard scanning:
[!] Missing API key for SecurityScorecard.
[*] Performing BuiltWith scan...
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
[!] Missing API key for BuiltWith.
┌──(kali㉿kali)-[~]
└─$


Nov 13 00:55
Zenmap
Scan Tools Profile Help
Target: 10.137.209.0/24
Profile: Ping scan
Scan
Cancel
Command: nmap -sn 10.137.209.0/24
Hosts
Services
OS Host
10.137.209.76
shashank-HP-La
_gateway (10.1
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
nmap -sn 10.137.209.0/24
Details
Starting Nmap 7.94SVN (https://nmap.org) at 2026-11-13 00:55 IST
Nmap scan report for _gateway (10.137.209.41)
Host is up (0.042s latency).
MAC Address: 92:A7:90:A9:E7:72 (Unknown)
Nmap scan report for 10.137.209.76
Host is up (0.17s latency).
MAC Address: 0A:27:32:C5:EA:3A (Unknown)
Nmap scan report for shashank-HP-Laptop-15s-gr0xxx (10.137.209.62)
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 5.37 seconds
Filter Hosts


Nov 13 01:42
Zenmap
Scan Tools Profile Help
Target: 10.137.209.0/24
Profile: Ping scan
Scan
Cancel
Command: nmap -sn 10.137.209.0/24
Hosts
Services
OS Host
10.137.209.76
shashank-HP-La
_gateway (10.1
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
Hosts Viewer
Fisheye
Controls
Legend
Save Graphic
10.137.209.76
shashank-HP-Laptop-15s-gr0xxx (10.137.209.62)
localhost
_gateway (10.137.209.41)
Filter Hosts

-End-

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📌 **Project Information**
Program Name: Cybersecurity program at Networkwalks | **Week:** 02 | **Repository:** https://github.com/shashankkbasurbasur-source/NETWORKWALKS-SHASHANK-BASUR-B084-WK2-PM4-PM5-CYBERSECURITY